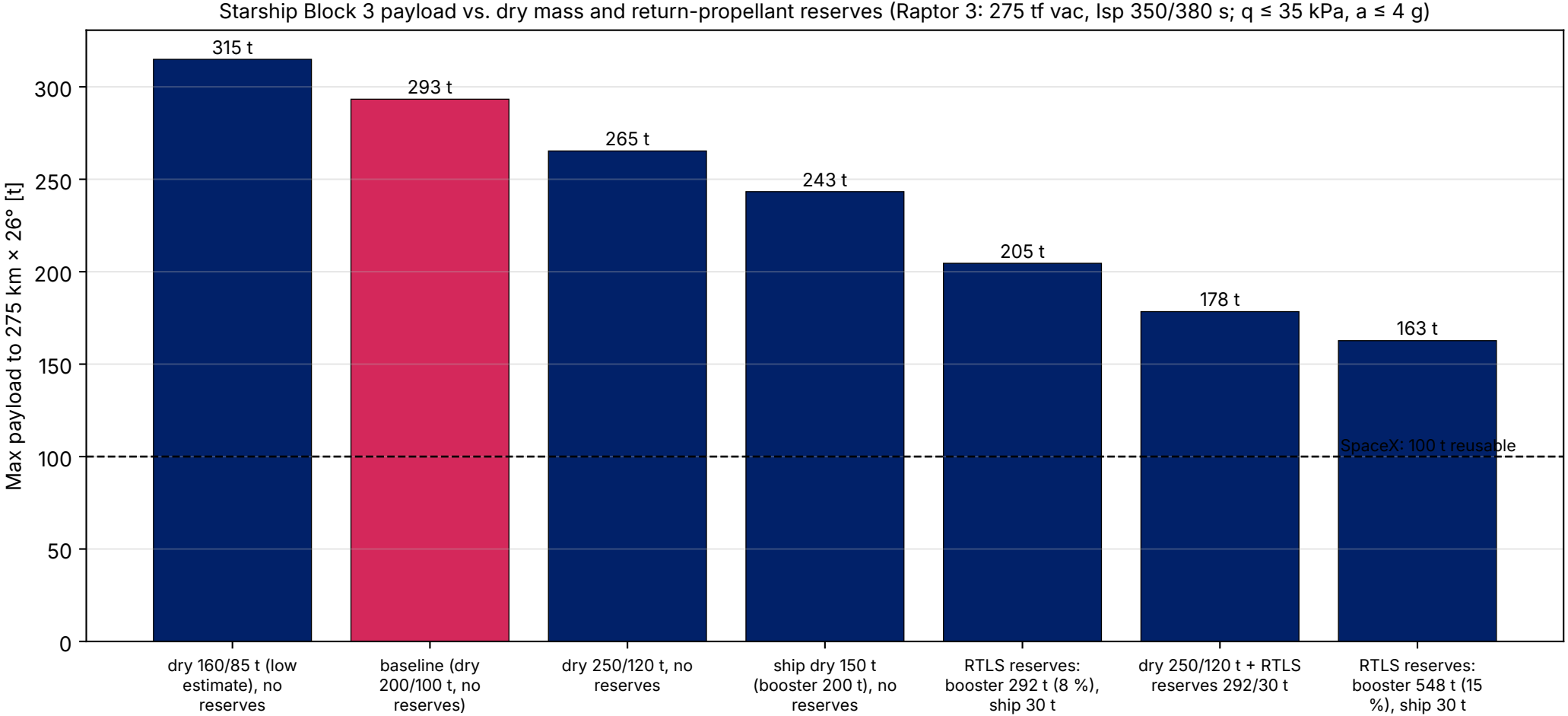
Starship Block 3 payload vs. dry mass and return-propellant reserves (Raptor 3: 275 tf vac, Isp 350/380 s; q ≤ 35 kPa, a ≤ 4 g)
Max payload to 275 km × 26° [t]
0
50
100
150
200
250
300
315 t
293 t
265 t
243 t
205 t
178 t
163 t
SpaceX: 100 t reusable
dry 160/85 t (low estimate), no reserves
baseline (dry 200/100 t, no reserves)
dry 250/120 t, no reserves
ship dry 150 t (booster 200 t), no reserves
RTLS reserves: booster 292 t (8 %), ship 30 t
dry 250/120 t + RTLS reserves 292/30 t
RTLS reserves: booster 548 t (15 %), ship 30 t